final_project_code

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1. Reading in the data

```
read_data <- read_excel("./stat571B_reading_data.xlsx")

read_data['total_time'] = read_data['Time (minutes)']*60 + read_data['Time (seconds)']

read_data['format'] = c("Laptop", "Laptop", "Print",
                        "Laptop", "Print", "Print",
                        "Print", "Print", "Laptop",
                        "Print", "Laptop", "Laptop",
                        "Laptop", "Print", "Print",
                        "Laptop", "Print", "Laptop",
                        "Print", "Laptop", "Laptop",
                        "Print", "Laptop", "Print",
                        "Laptop", "Print", "Laptop",
                        "Laptop", "Laptop", "Print",
                        "Print", "Laptop", "Print",
                        "Print", "Print", "Laptop")

read_data["music"] = c("no", "yes", "no",
                       "yes", "no", "yes",
                       "no", "yes", "no",
                       "yes", "no", "yes",
                       "no", "no", "yes",
                       "yes", "yes", "no",
                       "no", "no", "yes",
                       "yes", "yes", "no",
                       "no", "yes", "yes",
                       "yes", "no", "no",
                       "no", "yes", "yes",
                       "yes", "no", "no")

read_data
```

```
## # A tibble: 36 x 8
##    Participant  Text `Comprehension score (%)` `Time (minutes)` `Time (seconds)`
##    <chr>       <dbl>                     <dbl>            <dbl>            <dbl>
##  1 P1              1                       100                2               17
##  2 P1              2                        75                2               45
##  3 P1              3                        75                1               58
##  4 P2              1                        75                3               14
##  5 P2              2                        50                2               36
```

```
##  6 P2              3                        75                2                51
##  7 P3              1                        75                2                 5
##  8 P3              2                       100                2                28
##  9 P3              3                        75                1                49
## 10 P4              1                        75                2                22
## # i 26 more rows
## # i 3 more variables: total_time <dbl>, format <chr>, music <chr>
```

2. Analysis

We used ANOVA with participant as a blocking factor to assess the significance of the experimental factors, as required. Additionally, linear mixed-effects models were fitted with participant as a random effect to account for within-subject correlation.

We fit factorial models with format, music, and their interaction as primary fixed effects. Because participants each read multiple passages, participant was included to account for within-subject variability. Since only three text passages were used, Text was included as a fixed effect rather than a random effect to adjust for possible passage-specific differences.

Model 1: Time

```r
# ANOVA for Reading Time
model_time <- lm(total_time ~ format * music + Text + Participant + Participant:music,
                 data = read_data)
Anova(model_time, type = "III")
```

```
## Anova Table (Type III tests)
##
## Response: total_time
##                    Sum Sq Df  F value    Pr(>F)
## (Intercept)       31068.0  1 100.1084 8.454e-06 ***
## format              374.1  1   1.2054   0.30419
## music              1675.1  1   5.3975   0.04867 *
## Text               3617.3  2   5.8279   0.02744 *
## Participant       17428.3 11   5.1053   0.01437 *
## format:music          0.0  1   0.0001   0.99104
## music:Participant  7061.3 11   2.0685   0.15580
## Residuals          2482.8  8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
## Mixed Effects Model
model_time_lmer <- lmer(
  total_time ~ format * music +
    (1 | Text) +
    (1 + music | Participant),
  data = read_data
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
anova(model_time_lmer)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF  DenDF F value Pr(>F)
## format        340.4   340.4     1 21.932  0.7002 0.4117
```

```
## music        3854.8  3854.8     1 18.570  7.9289 0.0112 *
## format:music   12.0    12.0     1 21.932  0.0246 0.8768
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model 2: Comprehension

```r
# ANOVA for Comprehension
model_comp <- lm(`Comprehension score (%)` ~ format * music + Text +
                   Participant + Participant:music,
                  data = read_data)
Anova(model_comp, type = "III")
```

```
## Anova Table (Type III tests)
##
## Response: Comprehension score (%)
##                     Sum Sq Df F value    Pr(>F)
## (Intercept)       12604.2  1 52.3243 8.947e-05 ***
## format              833.3  1  3.4595   0.09994 .
## music               400.1  1  1.6609   0.23349
## Text                364.6  2  0.7568   0.50003
## Participant        3755.8 11  1.4174   0.31702
## format:music        416.7  1  1.7297   0.22489
## music:Participant  3281.2 11  1.2383   0.38986
## Residuals          1927.1  8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
## Mixed Effects Model
model_comp_lmer <- lmer(
  `Comprehension score (%)` ~ format * music +
    (1 | Text) +
    (1 + music | Participant),
  data = read_data
)
```

```
## boundary (singular) fit: see help('isSingular')
```

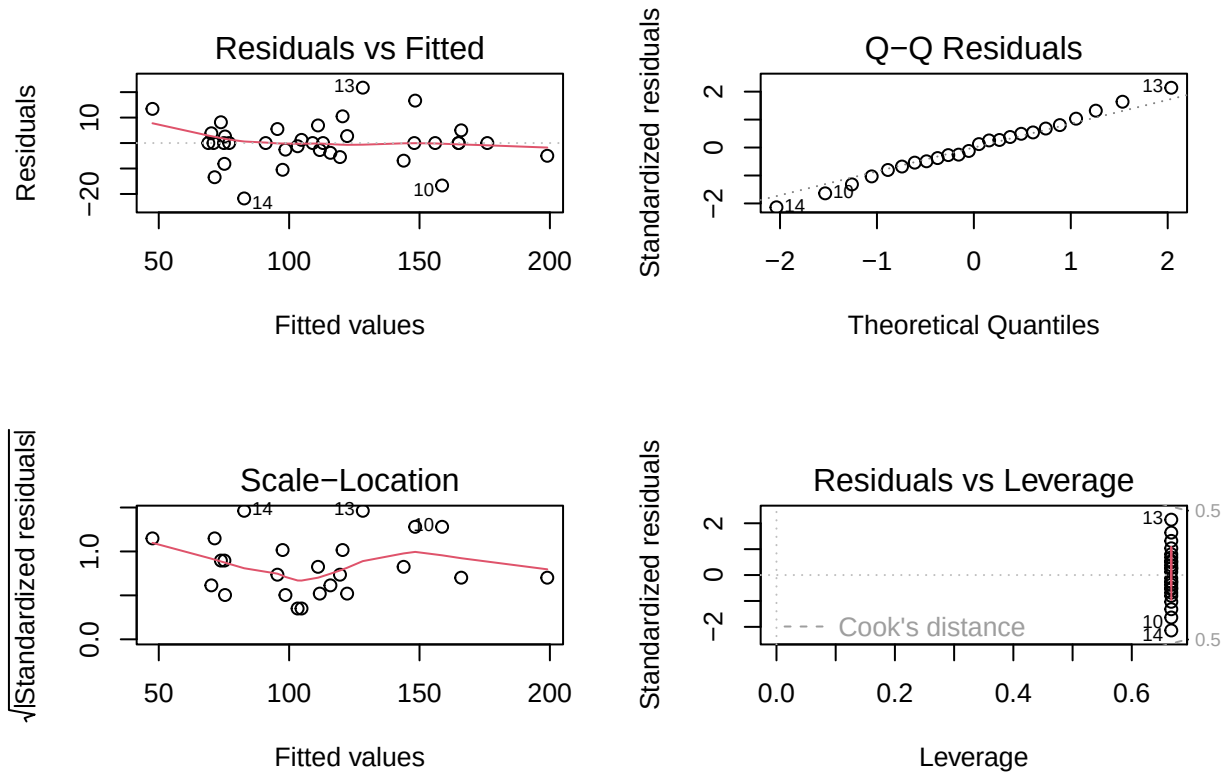
```r
anova(model_comp_lmer)
```

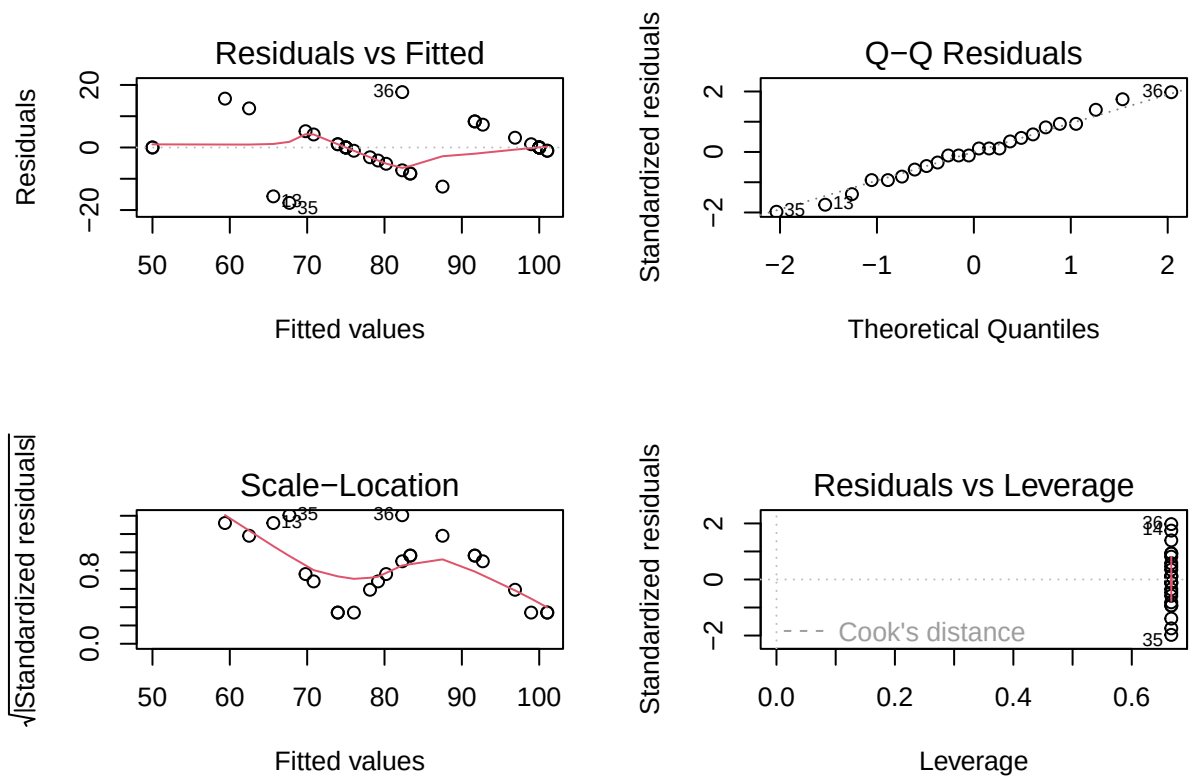
```
## Type III Analysis of Variance Table with Satterthwaite's method
##               Sum Sq Mean Sq NumDF  DenDF F value  Pr(>F)
## format         65.72   65.72     1 24.289  0.2764 0.60384
## music           1.28    1.28     1 20.002  0.0054 0.94230
## format:music 1145.62 1145.62     1 24.289  4.8179 0.03796 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Diagnostics for ANOVA

```r
par(mfrow = c(2,2))
plot(model_time)
```

```
## Warning: not plotting observations with leverage one:
##   2, 5, 8, 11, 15, 18, 21, 24, 25, 28, 31, 34
```




```r
par(mfrow = c(2,2))
plot(model_comp)
```

```
## Warning: not plotting observations with leverage one:
##   2, 5, 8, 11, 15, 18, 21, 24, 25, 28, 31, 34
```